

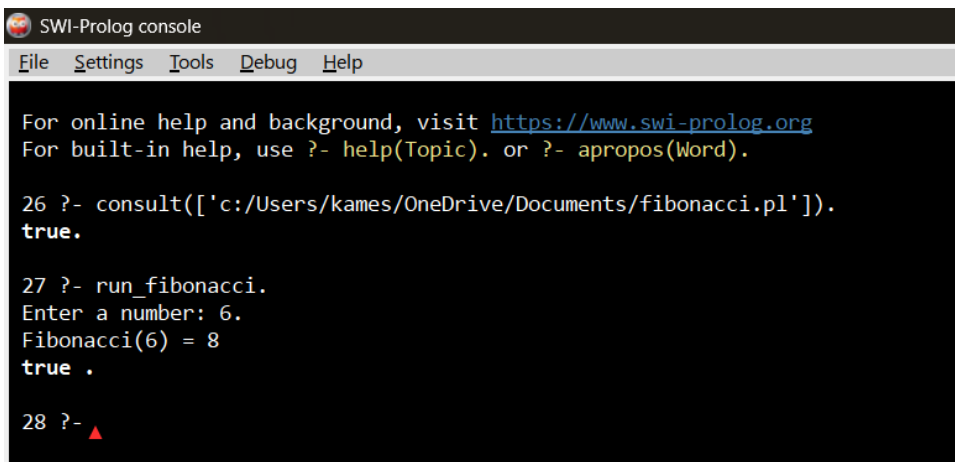
3-f Fibonacci in Prolog

PROGRAM:

```
fibonacci(0, 0).
fibonacci(1, 1).
fibonacci(N, R) :-
    N > 1,
    N1 is N - 1,
    N2 is N - 2,
    fibonacci(N1, R1),
    fibonacci(N2, R2),
    R is R1 + R2.

run_fibonacci :-
    write('Enter a number: '), read(N),
    fibonacci(N, R),
    format('Fibonacci(~w) = ~w~n', [N, R]).
```

OUTPUT:

A screenshot of the SWI-Prolog console window. The title bar says "SWI-Prolog console". The menu bar includes "File", "Settings", "Tools", "Debug", and "Help". The console text shows instructions for online and built-in help. It then shows the execution of the program: a Prolog query to consult the file 'c:/Users/kames/OneDrive/Documents/fibonacci.pl' returns 'true.'. Next, the 'run_fibonacci' predicate is called, which prompts for a number. The user enters '6', and the console displays 'Fibonacci(6) = 8' followed by 'true.'. The prompt '28 ?-' is shown at the bottom with a red triangle cursor.

```
SWI-Prolog console
File Settings Tools Debug Help

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

26 ?- consult(['c:/Users/kames/OneDrive/Documents/fibonacci.pl']).
true.

27 ?- run_fibonacci.
Enter a number: 6.
Fibonacci(6) = 8
true .

28 ?- ▲
```